DATA ANALYTICS INTERNSHIP – TASK 3

DEEP-DIVE ANALYSIS & INTERACTIVE DASHBOARDING

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Internship: Data Analytics Internship

Tool Used: Power BI Desktop

Task: Deep-Dive Analysis & Interactive Dashboarding

1. **Objective**

Evaluate regional sales distribution.

Explore product rankings.

The objective of this project is to perform a detailed analysis of sales data and build an interactive Power BI dashboard that helps users understand business performance through visual analytics.

The dashboard enables users to:

Monitor sales performance.

Analyze profit trends.

Compare category-wise performance.

Apply interactive filters for deeper analysis.

2. Dataset Overview

The dataset contains sales transaction records including:

- Order ID
- Order Date
- Product
- Category
- Region
- Sales
- Profit

The data was imported into Power BI and used for creating KPIs, charts, and interactive dashboard pages.

3. Key Performance Indicators (KPIs)

The following KPIs were created:

KPI	Value
Total Orders	40
Total Sales	510K
Total Profit	104K

These KPIs provide a quick overview of business performance.

4. Dashboard Page 1 – Sales Performance Dashboard

Dashboard Components

KPI Cards

- Total Orders
- Total Sales
- Total Profit

Charts Used

- Treemap – Category Wise Profit
- Line Chart – Monthly Sales Trend
- Bar Chart – Region Wise Sales
- Pie Chart – Category Wise Sales
- Gauge Chart – Sales Target Achievement
- Product Wise Sales Chart

- Profit Distribution by Region

Slicers Used

- Category Filter
- Region Filter
- Date Filter

Purpose

This page provides a high-level overview of sales and profit performance. Users can quickly identify top-performing categories, regions, and products.

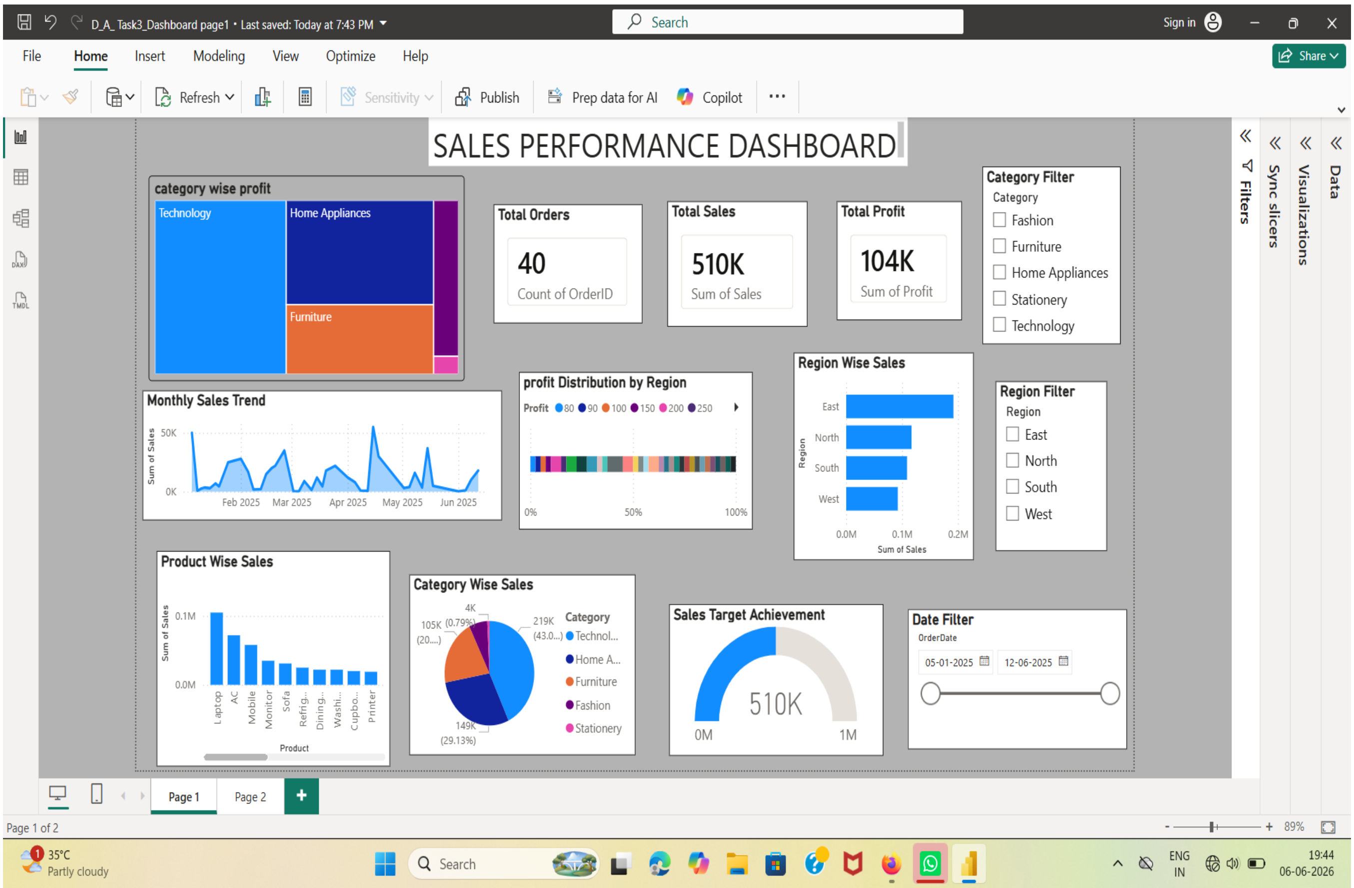
Key Findings

Technology category generated the highest sales and profit.

East region recorded the highest sales performance.

Monthly sales showed fluctuations across the year.

Overall sales reached approximately 510K.



5. Dashboard Page 2 – Advanced Sales Analysis

Dashboard Components

<u>Charts Used</u>

- Region Vs Category Analysis Matrix
- Sales vs Profit Scatter Chart
- Product sales Ranking Funnel Chart
- Product Ranking Over Time

- Sales by Category Doughnut Chart
- Profit by Product Analysis

Slicers Used

- Category Filter
- Region Filter
- Date Filter

Purpose

This page provides deeper insights into sales and profitability across products, categories, and regions.

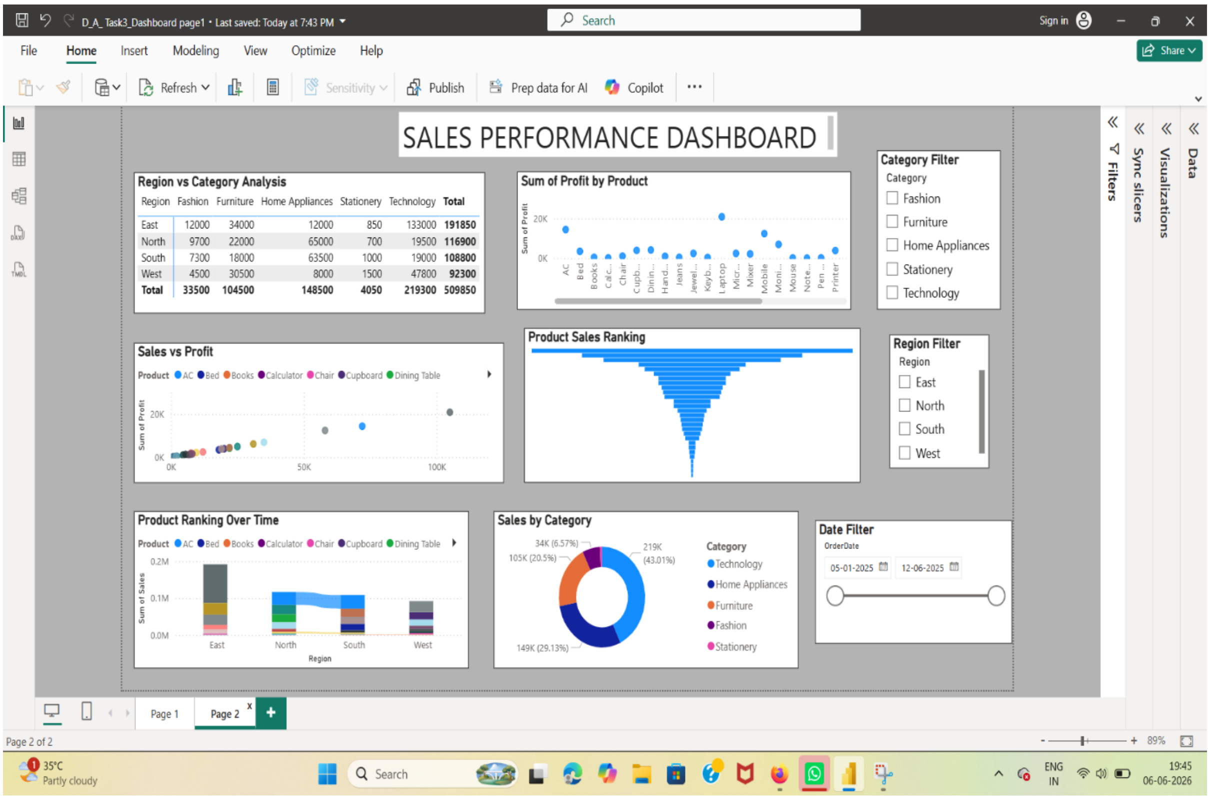
Key Findings

Certain products generated significantly higher profits.

Strong positive relationship observed between sales and profit.

Technology products dominated sales rankings.

Category performance varied significantly across regions.



6. Interactive Features

The dashboard includes:

Dynamic filtering using slicers.

Cross-filtering between visuals.

Interactive drill-down analysis.

Real-time KPI updates based on user selections.

These features improve user experience and support decision-making.

7. Business Insights

1. Technology category is the strongest contributor to revenue.

2. East region consistently performs better than other regions.

3. High-sales products contribute significantly to overall profit.

4. Interactive analysis helps identify business opportunities quickly.

5. Dashboard provides actionable insights for management decisions.

8. Conclusion

This project successfully demonstrates the use of Power BI for business intelligence and dashboard development. The interactive dashboard provides meaningful insights into sales performance, profit trends, regional analysis, and category performance.

The developed dashboard enables users to monitor KPIs, analyze business performance, and make data-driven decisions effectively.

Tools Used

- Power BI Desktop
- CSV Dataset
- Data Visualization Techniques